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Review

A Systematic Review on Research Documentation of Ocular Medical Device focused on Perfluorocarbon Liquid

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Abstract

Medical devices are used worldwide and a good annual consumption of ocular medical devices are reported. Medical devices are being used in healthcare domain for diagnosis, monitoring, prevention and treatment of an array of diseases. The use of medical devices benefits patients immensely, but they also carry significant potential risk. Recently, countries like- Spain, Holland, France, Italy, Middle East, Switzerland and Chile have reported several cases of acute retinal toxicity due to perfluorocarbon liquids (PFCL), especially perfluoro-octane (PFO). These reported cases cause a panic situation among healthcare practitioners such as doctors, nurses, technicians, biomedical engineers and medical device stakeholders. Therefore, the regulations & scientific publications of such medical devices across globe are the need of an hour. Global regulation on PFCL was thoroughly studied and analyzed. The web of science-based search was performed on scientific publications available worldwide. This paper reviews the global regulation & scientific publications on PFCL, thus, provides an idea about the need of further improvement.

Keywords: Medical device, regulation, PFCL toxicity, adverse event.

Introduction

Today, there are an estimated 2 million different kinds of medical devices used in the global market, categorized into more than 7000 generic devices groups [1]. Over the last many years, medical devices have been playing a pivotal role in the diagnosis and management of a variety of diseases [2]. However, these devices have also created substantial risks to the patients with high

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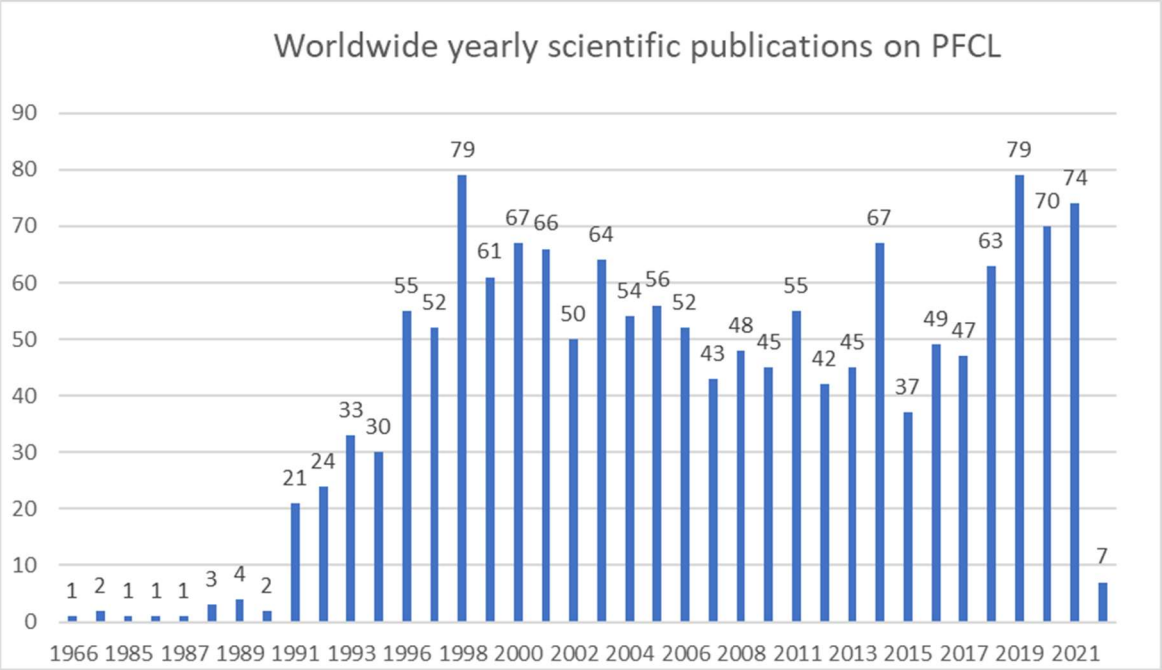


Figure 1 *Worldwide yearly scientific publications on Perfluorocarbon liquid.*

profile recalls. Nearly 5,000 individual classes of medical devices, tens of thousands of medical device suppliers, and millions of healthcare providers exist worldwide, which clearly depicts that they also carry significant risk potential [3]. Countries like- Spain, Holland, France, Italy, Middle East and South America have reported several cases of acute retinal toxicity due to perfluorocarbon liquids (PFCL), especially perfluoro-octane (PFO). As per WHO, medical devices including an instrument, apparatus, appliance, implant, material or other article, whether used alone or in combination, including a software or an accessory, intended by its manufacturer to be used specially for human beings which does not achieve the primary intended action in or on human body or animals by any pharmacological or immunological or metabolic means, but which may assist in its intended function by such means for one or more of the specific purposes of - diagnosis, prevention, monitoring, treatment or alleviation of any disease or disorder; diagnosis, monitoring, treatment, alleviation or assistance for, any injury or disability; investigation, replacement or modification or support of the anatomy or of a physiological process; supporting or sustaining life; disinfection of medical devices; and control of conception [4]. On comparing, definition of medical devices with United State Food and Drug Administration (USFDA) & European Union (EU), it was observed that the definition of USFDA clearly illustrates that medical devices achieve their effect via physical or mechanical properties and are often chemically and biologically inert [5]. In EU definition, it clearly illustrates that medical devices do not attain their action by pharmacological, immunological or metabolic means, but may assist in its function by such means [6]. Medical device regulation & definition

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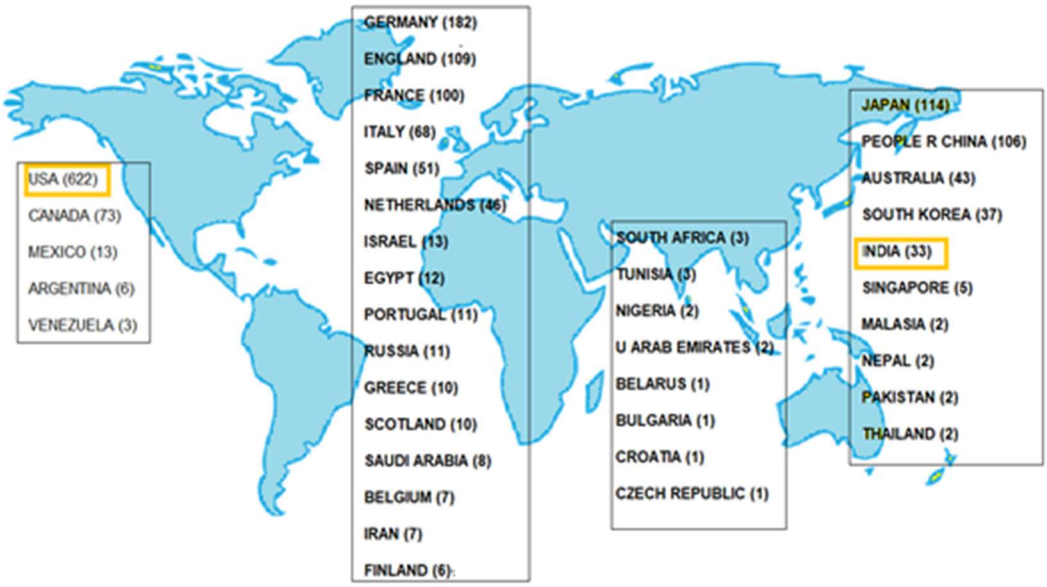


Figure 2 *Worldwide countries publishing documents related to Perfluorocarbon liquid.*

across the world is very varied, ranging from comprehensive to none. It is necessary to promote global harmonization on medical device regulation and dissemination of medical device safety information among different regulatory authorities. The purpose of this review is to compile the current global situation related to PFCL and focus on the need for an appropriate and more robust medical device regulation. More scientific publications focused on the safety of PFCL are required to aware the healthcare professionals and it helps to improve patient safety.

Global regulation on PFCL

Medical devices have been used for medical purposes from many years in several countries. Each country or region has its own regulatory authority such as European Medicines Agency (EMA) for European region, Food and Drug Agency (FDA) for USA and Pharmaceuticals and Medical Devices Agency (PMDA) for Japan [7] are some of them. These principal agencies or their appointed agencies such as notified bodies in the case of Europe are overseeing the certifications, marketing and adverse effects of medical devices in their own territories. Similar is the case for other countries. Most of them have their own regulatory standards and guidelines for medical devices. Although there are collaborations worldwide among the regulatory authorities, they still lack to produce a global harmonization for regulatory requirements, definition, classes and subclasses of medical devices. In Europe, there are three main classes to differentiate among medical devices: Class I, II and III following the classification rules of Annex IX of the Medical Device Directive (MDD) 93/42/EEC [8]. Class I is the lowest risk and class III is the highest risk class of medical devices. Class II has two subclasses: IIa for medium risk and IIb for medium-to-high risk class of medical devices. Each medical device is categorized in one of these four mentioned classes. PFCL are classified under class IIb

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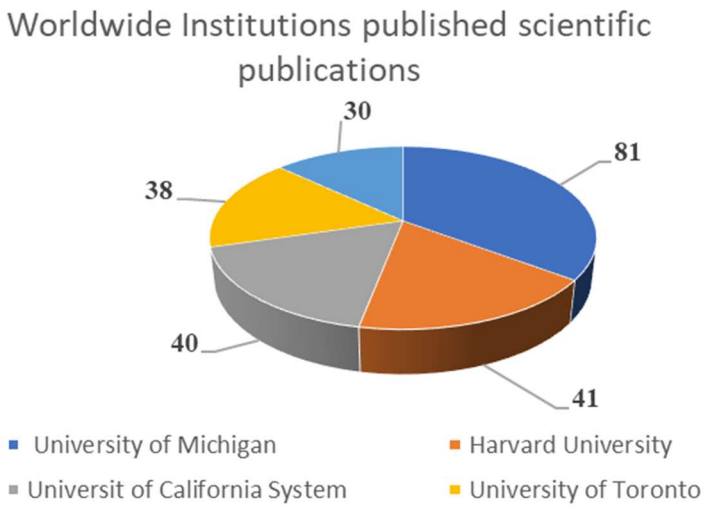


Figure 3 *Worldwide institutions published scientific publications on Perfluorocarbon liquid.*

(medium to high risk). Currently, Europe has 50 notified bodies for enforcing current MDDs [9]. National competent authorities of European member states assign testing laboratories as notified bodies (NB), which perform tasks under their monitor and audit. These independent NB provide auditing services for medical devices including medical device certification such as

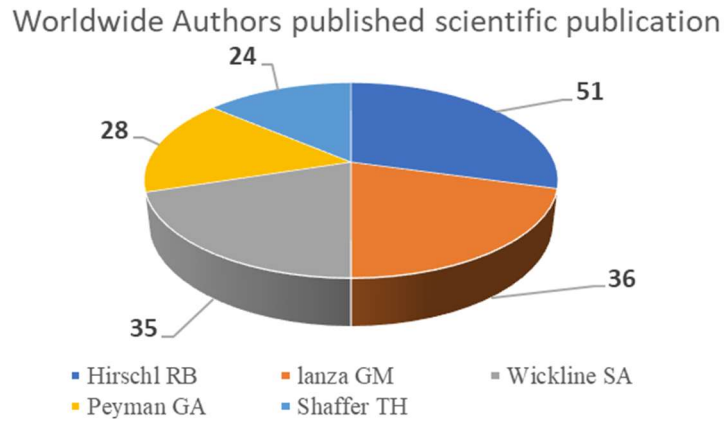


Figure 4 *Worldwide authors published scientific publication on Perfluorocarbon liquid.*

recommendation to assigning Conformité Européenne (CE) mark for Europe or United Kingdom Conformity Assessed (UKCA) mark for UK, device type designation, quality systems assessments and verifications and review of design dossiers for high-risk devices [10]. A manufacturer can select a notified body dealing to one of these three classes that is considered relevant to its fabricated medical device for passing the rigorous pre-market approval process to obtain CE certification. European laws impose obligations on manufacturers to actively monitoring and reporting the incidents associated to medical device to national authorities (known as vigilance system), but these laws are unhelpfully vague and little detail is given on how diligently the manufacturer should try to find out about such incidents [11]. Recent public health safety issues involving medical devices have led to a growing demand to improve the current passive-reactive post-market surveillance (PMS) system [12]. Currently the new regulations in Europe require each manufacture proactively monitoring the benefit/risk ratio of an approved medical device during its entire life cycle. It includes collecting and reporting to member state via the competent authority the post-market surveillance (PMS) data associated to adverse events of each medical device. In United States of America (USA), Manufacturer and User Facility Device Experience (MAUDE) database, UK, Medical and Healthcare Products Regulatory Agency (MHRA) database and Canada, Health Canada database are established databases

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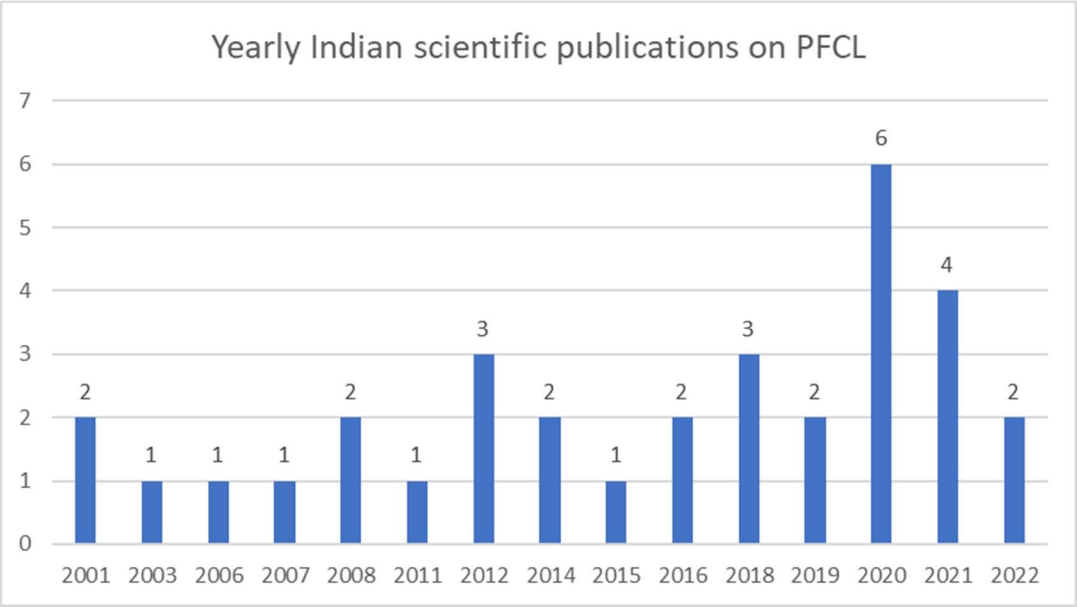


Figure 3 *Yearly Indian scientific publications on Perfluorocarbon liquid.*

where the adverse events are directly reported, or end-users can report via manufactures to relevant national authorities.

United States regulates the medical devices via FDA, specifically through the Center for Devices and Radiological Health (CDRH) [13]. Devices are classified according to their perceived risk using a 3-tiered system (class I, II, or III). Class I is for lowest risk; Class II is for higher-risk devices and Class III is for highest potential risk. PFCL are classified under class III (highest potential risk). The manufacturers need Premarket Notification or Premarket Approval (PMA) for commercially distributing a medical device in USA territory. Most Class I devices are exempt from Premarket Notification. Besides, an Investigational Device Exemption (IDE) allows obtaining FDA approval for clinical testing of an unapproved significant-risk medical device. In addition to FDA, the Institutional Review Board (IRB) must review and approve the study before it starts at the clinical site. The IDE application provides information to the FDA on device design and qualification, as well as on the study protocol. Thus, IDE allows manufacturers to collect medical device related safety and effectiveness data in clinical trial set-up, which is submitted to FDA for PMA or Premarket notification. Three different trials categories are under this IDE process: noninferiority, equivalency, and superiority [14]. The applied IDE regulation for a medical device must be justified and if there is a source of additional information, it must be listed in justification for obtaining IDE approval.

African countries, mostly, lack their own structures of authorities for regulating the medical devices and their adverse effects. They normally follow the European or FDA or in some cases, their own approved regulations and classification systems for medical devices, confirming again the lack of global harmonization in this regard. For example, Egyptian Drug

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Authority (EDA) follows the European medical classification system (Class I, IIa, IIb and III) and regulations.

In South Africa, Department of Health, South Africa Medicines Control Council (MCC) follows the No. 101/1965, No. 14/2015 and No. 15/1973 and medical devices are classified in A, B, C and D class. Their classification is in line with Indian classification of medical devices. The Nigerian National Agency for Food and Drug Administration and Control (NAFDAC) recommends the compliance to classification of the country where the medical device is manufactured and follow the Cap N1 L.F.N 2004 [15]. The Food Medicine and Healthcare Administration and Control Authority of Ethiopia (FMHACA) of Ethiopia classifies the medical devices in Class I, II, III, IV and follow No. 661/2009 and follow No. 12/2013 and No. 9/2012. It should be noted that vast African population do not have access to either medical devices or a government approved public authorities and structures to report if there is any adverse effect due to a medical device. However, most of these medical devices are manufactured in USA, Europe, India, China or any other foreign country. Undeveloped globally coordinated structure inhibits inter-country communications and research publications.

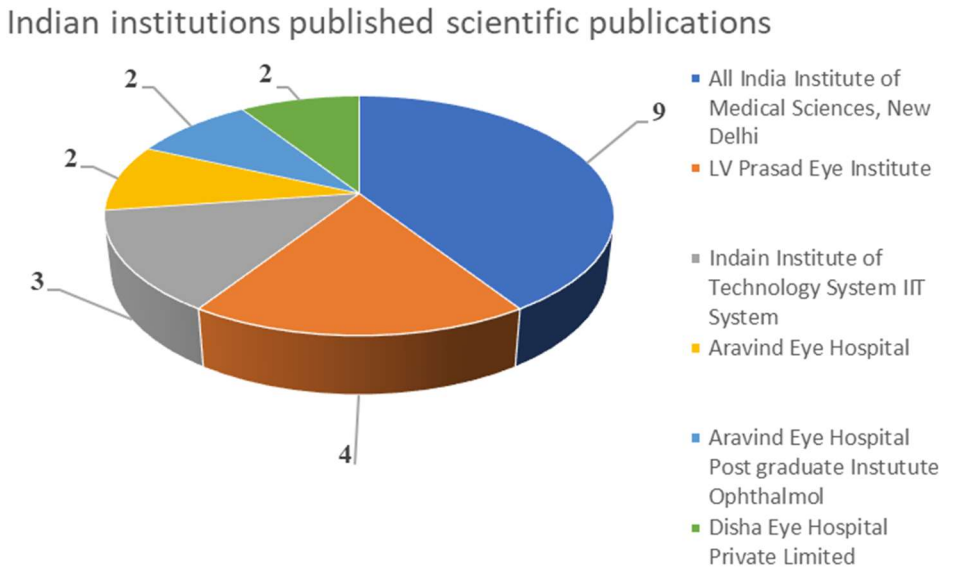


Figure 4 *Indian institutions publishing scientific publications on Perfluorocarbon liquid.*

Indian regulation on PFCL

In India, Central Drugs Standard Control Organization (CDSCO) is the national regulatory authority under Directorate General of Health Services of Ministry of Health & Family Welfare of Government of India. All the Medical devices are regulated by the medical device rules (MDR), 2017 [16]. Provisions related to import, manufacture, labelling, distribution and sale of medical devices including in-vitro Diagnostics (IVDs) are covered under MDR, 2017. Government of India has notified these rules on January 01, 2017, vide General Statutory Rules (GSR) 78 E and it is effective since January 1, 2018. According to MDR, based on the risk, medical devices are classified into four categories- Class A, B, C & D. Class A for low risk, Class B for low to moderate risk, Class C for moderate to high risk & Class D for

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high risk. PFCLs are classified under Class D (high risk) [17].

Global Scientific Publication analysis on PFCL

Bibliography analysis

A search has been performed on date March 01, 2022, in Web of Science to know how many research papers have been published using the terminology "Perfluorocarbon liquid". A total of 1,628 results from Web of Science Core Collection is found from 1966 to 2021. Searched publications includes reviews, early access publication and open access publication. It is quite interesting to note that after 2018, in the last 3 years (2019, 2020 and 2021), more than 70 research documents are published. This observation has not been seen from 1966 until 2018 except 1998 when 79 research documents are published (Figure 1). This clearly suggests that the scientific community including healthcare practitioners are reporting more findings in the last 3 years.

If countries are compared for publishing PFCL related scientific publications, it is found that USA has published 622, Germany- 182, Japan- 114, England- 109, China- 106 and France- 100 respectively. Rest of the countries has published less than 100 scientific publications including India has published only 33 documents (Figure 2).

Similarly, it is observed that most research documents are published from the University of Michigan of USA (81 scientific publications). Rest of research institutions are also american based and published 41 or less scientific publications (Figure 3). Out of 81, 51 was published by Prof Hirschl RB of the University of Michigan. But when it is filtered by using the term "Ophthalmology", it showed that Prof Peyman GA of the Louisiana State University of USA has published 28 scientific publications (Figure 4). The rest of authors of ophthalmology field have published less than 7 research documents.

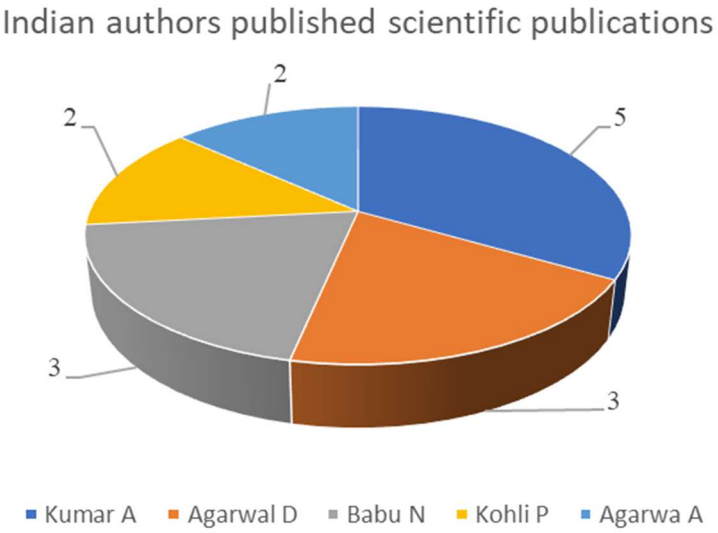


Figure 5 *Authors of India publishing scientific publication on Perfluorocarbon liquid.*

If we select "India" as a country/region for published research works or case reports, a total of 33 results has been found. Searched results includes review articles, early access article and open access literatures. It is quite interesting that the first scientific publication published in 2001 and a maximum of six published in 2020 (Figure 5). In India, majority of research documents are published from the All India Institute of Medical

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Sciences (AIMS), New Delhi (Figure 6). Out of 33, 5 scientific publications are published by author Kumar A (Figure 7).

Global scenario on adverse event reporting on PFCL

PFCL are available in the market as Perfluoro-octane (PFO), Perfluorodecaline (PFD) etc but mostly, in ophthalmology, they are used in retinal manipulation and several ocular surgical procedures due to their optimal properties. For example, PFO is a widely used intraoperative ocular medical device, used extensively in ocular surgeries such as retinal detachments (RD), intra ocular lens (IOLs) dislocation, and as giant retinal tears (GRT). Nevertheless, PFCL are chemically inert but have significant physical properties, allowing their applications as a heat transfer agent, dielectric fluid and tamponade in eye surgery. As an ocular endotemponade agent, PFCL, due to reason of being chemically inert and without any biological and therapeutic activity, support them to be classified as a medical device. They are also referred as “heavy liquid” due to their high specific gravity which supports their ability to flattens the folds or tears in retina which eases the surgical operation, and the high interfacial tension and low surface tension allow administration of this heavy liquid in the vitreous cavity in form of a bubble limiting the spread and ensuring the ease in administration and removal of the liquid. The low viscosity further eases the administration and removal of this liquid, and the optical clarity of this liquid provides the maneuverability to the surgeon. These properties make PFCL an excellent ocular intraoperative tool for retinal manipulation and several ocular surgical procedures such as RD, RD in proliferative vitreoretinopathy (PVR), GRT, treatment of macular hole, dislocated lens, foreign body removal, silicon oil removal, to drain suprachoroidal hemorrhage, in treatment of proliferative diabetic retinopathy, etc. Recently countries like- Spain, Holland, France, Italy, Middle East, Switzerland and Chile have reported several cases of acute retinal toxicity due to PFCL, especially PFO. Spain has reported 128 cases, Switzerland has reported 48 cases and Chile has reported 14 cases of blindness after intraocular surgery due to acute toxicity of PFO [18]. These reported cases cause a panic situation among healthcare practitioners such as doctors, nurses, technicians, biomedical engineers and medical device stakeholders. Worldwide, continuous monitoring is required on the usage of PFCL as well as manufactures should follow the quality standard in order to ensure the safety & efficacy of PFCL in clinical use. It is necessary to publish more scientific publications and disseminate them to healthcare practitioners globally.

Indian scenario and mechanism of adverse event reporting on PFCL

Medical devices are being used in healthcare domain for diagnosis, monitoring, prevention and treatment of an array of diseases. The use of medical devices benefits the patients immensely, but they also carry significant risk potential. To

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ensure the safety of medical devices, ministry of health & family welfare, government of India has approved a commencement of "Materiovigilance Programme of India (MvPI)" on February 10, 2015. Materiovigilance programme was launched on July 06, 2015, at Indian Pharmacopoeia Commission (IPC), Ghaziabad by Drugs Controller General of India. IPC functions as a National Coordination Centre for this programme [19]. The aim of this programme is to identify the adverse events associated with the use of medical devices and to eliminate the device-related risks through a systematic reporting system. Under MvPI, various modalities to report adverse events associated with medical devices have been developed. These modalities include an editable medical device adverse event reporting form, a toll-free helpline number (18001803024) and a field safety corrective action form to notify the regulatory authority and healthcare professionals on corrective actions or recall by the manufacturer [20]. In quo of Coronavirus disease 2019 (COVID-19) pandemic, one-page editable form has been developed to boost the adverse event reporting of Personal Protective Equipments (PPEs). NCC-MvPI, IPC receives medical device adverse event reports through 293 medical device adverse event monitoring centres (MDMCs), 652 adverse drug reaction monitoring centres (AMCs) [centres established under Pharmacovigilance Programme of India (PvPI)], Marketing authorization holders (MAHs) and consumers. MvPI collects reports and registers in MvPI database. Preliminary examination is done by MvPI team in order to ensure the completeness of reported cases and secondary examination is done pool of subject experts of different departments (cardiology, gynecology, orthopedics, urology, etc.), thus team receive their independent opinions on the reported cases. MvPI subject expert helps in identifying the causality of reported cases. The outcome received from subject experts is discussed in MvPI technical meeting and final findings in terms of safety alerts & recommendations. Safety alerts are circulated to monitoring centres for the safety surveillance of medical devices and recommendations are forwarded to CDSCO for their information and necessary actions at their end. In MvPI database, there are no adverse event associated with PFCL is reported.

Conclusion

Monitoring and regulations on PFCL must be improved worldwide, and a continuous monitoring is required on the usage of PFCL by the healthcare practitioners and consumers. It is necessary to report adverse events due to PFCL use, publish more scientific publications and disseminate them to healthcare practitioners globally.

Conflict of interest

Authors declare no conflict of interest.

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